Well Killer User Manual

LoRa Irrigation Well Control System
User Documentation

1. Introduction

Well Killer is a C++ software simulation for a LoRa-based irrigation well monitoring and emergency shutdown system. It is designed to let a farmer view well status, remotely stop an individual well, stop all wells during an emergency, review shutdown events, and test simulated LoRa communication.

This version simulates the software, LoRa transmitter, LoRa receiver, and Murphy shutdown circuit. It does not directly control real equipment.

2. Starting the Program

Open the WellKiller project in Microsoft Visual Studio, build the solution, and run the application. The opening screen explains the purpose of the system and then displays the main menu.

The program creates seven simulated irrigation wells. Each well begins in the RUNNING state.

3. Main Menu

The main menu contains six options:

Option	Function
1	Display Well Status
2	Shut Down One Well
3	Shut Down All Wells
4	View Shutdown Log
5	Test LoRa Communication
6	Exit

4. Display Well Status

Choose option 1 to display all seven wells and their current status. A well will be shown as RUNNING or STOPPED.

5. Shut Down One Well

Choose option 2 to display the wells and enter the number of the well that should be stopped.

The program simulates the shutdown sequence by sending a LoRa command, receiving the command at the well, triggering the Murphy shutdown circuit, changing the well status to STOPPED, and recording the event in the shutdown log.

If the selected well is already stopped, the program reports that no additional shutdown is required.

6. Emergency Shut Down All Wells

Choose option 3 when all operating wells need to be stopped. The program displays a warning and asks for confirmation.

Enter Y to continue. Any well that is still running will go through the shutdown sequence. Enter N or another response to cancel the operation.

7. View Shutdown Log

Choose option 4 to display shutdown events stored in WellKillerLog.txt.

If no shutdown events have been recorded yet, the program reports that the log is empty.

8. Test LoRa Communication

Choose option 5 to run the simulated communication test. The transmitter sends a test signal and the receiver reports that the signal was received. A successful simulation ends with a LoRa communication test PASSED message.

9. Exit

Choose option 6 to exit the application. The program displays a safe shutdown message before closing.

10. Input and Error Messages

Menu choices must be numbers from 1 through 6. Well numbers must also be numeric and must identify an existing well.

INPUT ERROR indicates invalid input such as entering text where a number is expected. RANGE ERROR indicates a number outside the allowed range. SYSTEM ERROR is used for other application errors, such as failure to open the shutdown log file.

11. Safety and Simulation Notice

This software is currently a simulation and should not be treated as a certified industrial safety control. Real hardware deployment would require appropriate electrical protection, fail-safe design, testing, and verification of the LoRa and Murphy shutdown interfaces.

Quick Reference

• Normal monitoring: Select 1.

• Stop one well: Select 2, then enter the well number.

• Emergency stop: Select 3, then confirm with Y.

• Check history: Select 4.

• Check simulated radio link: Select 5.

• Close program: Select 6.